12-LEAD ECG IMAGE-BASED PREDICTION OF LEFT VENTRICULAR EJECTION FRACTION FOR EARLY HEART FAILURE DETECTION USING DEEP LEARNING

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in Electronics and Communication Engineering

by

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APPROVAL OF THE PROJECT EVALUATION BOARD

19/11/2026

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Date: 9th November 2026

DECLARATION

We certify that

a. The work contained in the report is original and has been done by us under the general supervision of our supervisor(s).

b. The work has not been submitted to any other Institute for any degree or diploma.

c. We have followed the guidelines provided by the Institute in writing the report.

d. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references.

e. Whenever we have quoted written materials from other sources, we have put them under quotation marks and given due credit to the sources by citing them and giving required details in the references.

f. We took help from an AI assistant for some tasks, such as writing and checking code, running the documented experiments and checking the wording. All design decisions, data handling approvals and conclusions are our own.

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Abstract

Heart failure with reduced left ventricular ejection fraction (LVEF) is common and treatable but often detected late, because echocardiography is not available at every point of care. The 12-lead ECG is recorded routinely, yet in many Indian hospitals it survives only as a printed PDF page. This project grades LVEF into four classes (Normal, Mild, Moderate, Severe) from such printout images alone.

The data are 1,047 single-centre printouts from 993 patients, labelled from echocardiography: 837 ECGs for development and 210 kept apart as a locked test set. The traces were recovered exactly from the vector PDFs. Every model was evaluated with patient-grouped, repeated, nested cross-validation, corrected intervals and a written pre-registration, across ten experimental stages covering image CNNs, the ECGFounder foundation model, 594 engineered features, gradient boosting, staged and ordinal ensembles and triage prediction. The public EchoNext data (4,827 ECG–echocardiography pairs of all EF classes) were redrawn in the hospital format and used for transfer learning.

Every model was trained, for as many epochs as it needed, until its training accuracy reached (nearly) 100 %; 81 such models are reported. The best four-class accuracy on all patients, from image-derived inputs only, is 0.558 [0.530, 0.587], by an ensemble that includes EchoNext-trained members; the best single model reaches 0.557 and the best model without the printed measurements 0.549. A triage model that answers only confident cases is correct for 0.696 [0.630, 0.762] of the 48 % of patients it answers, and on clean, clear-cut labels the models reach 0.67–0.69. Models trained only on EchoNext reach up to 0.539 on the hospital printouts with only their thresholds tuned. Although every flexible model fits its training data perfectly, held-out accuracy stays at 0.35–0.56. A simulation explains the ceiling: the ECG-based EF estimate correlates with echo at $r \approx 0.62$, and 0.70 on all patients would need $r \approx 0.90$.

Phase II will use digital ECGs, larger linked datasets and verified Simpson-method EF labels.

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List of Abbreviations

AUROC	Area under the receiver operating characteristic curve
CI	Confidence interval (corrected for repeated cross-validation unless stated)
CNN	Convolutional neural network
CV	Cross-validation
ECG	Electrocardiogram
ECGF, ECGFounder	ECG foundation model pre-trained on more than 10 million ECGs (Li et al.)
EchoNext	Public ECG–echocardiography dataset (PhysioNet), about 100,000 ECGs
EF, LVEF	(Left ventricular) ejection fraction, in per cent
GBM	Gradient-boosting machine (LightGBM)
HF	Heart failure
LR	Logistic regression
Mi, Mo, N, S	Mild, Moderate, Normal and Severe classes
MLP	Multi-layer perceptron
STAGED-4	Staged four-class decision: Normal / {Mild, Moderate} / Severe, then Mild vs Moderate
SUPER	Ensemble of all image-derived members pruned inside each training fold
TCRT	Total cosine R-to-T
TOP10	Ensemble keeping the ten best single members per fold

Notation

y	Class label $y \in \{0, 1, 2, 3\}$: Normal, Mild, Moderate, Severe
$\hat{\mu}$	Predicted EF (per cent)
σ	Standard deviation of the EF estimation error
r	Pearson correlation between predicted and echo EF
$\Phi(\cdot)$	Standard normal cumulative distribution function
$\mathbf{w}$	Class weights of the decision rule
J	Number of cross-validation cells (here 15)

Chapter 1

Introduction

1.1 Background and motivation

Heart failure affects tens of millions of people, and its outcome depends on how early left ventricular (LV) systolic dysfunction is recognised [1, 2]. Systolic function is graded by the left ventricular ejection fraction (LVEF), measured by echocardiography, which needs an ultrasound machine and a trained operator. The 12-lead electrocardiogram (ECG), in contrast, is recorded for almost every cardiac patient, and deep learning has shown that it carries information about reduced LVEF [3, 4]. That evidence comes mostly from digital signals and very large datasets. In many Indian hospitals the ECG survives only as a printed page or its PDF, which shows each lead for just 2.5 s plus one 10-s rhythm strip.

1.2 Problem statement

Given only the image of a hospital 12-lead ECG printout, predict the patient's four-class LVEF category:

- **Normal:** EF above 52 % (men) or at least 54 % (women) [5];
- **Mild:** 41 % up to the Normal limit;
- **Moderate:** 31 to below 41 %;
- **Severe:** below 31 %.

The model may use only the waveform on the page: no printed machine measurements, age or sex. The Mild band, only 11–13 EF points wide, is where an early warning is most valuable and where the ECG changes are subtle. A useful model must therefore detect Mild and not only the extremes.

1.3 Objectives

1. Prepare a labelled ECG image dataset: clean the ECG and echocardiography records, link every ECG to its patient's echocardiogram, extract the LVEF and assign one of the four classes.
2. Recover the exact ECG signal from the printout and build image-derived representations of it.
3. Train and compare the widest practical range of models, from classical features and CNNs to foundation models and ensembles, under a nested and preregistered protocol.
4. Transfer knowledge from public ECG–echo data (EchoNext) drawn in the hospital printout format.
5. Find the best all-patient four-class model and a triage model that answers confidently.
6. Train every model until it fits its training data (nearly) perfectly, measure its accuracy on new patients, and quantify the accuracy that the data can support.

1.4 Contributions

- A Python data-preparation pipeline: cleaning of the ECG and echocardiography spreadsheets, fuzzy matching of the two record sets, LVEF label generation and class-wise organisation of the ECG files.
- An exact vector digitiser and an inverse printout renderer for the hospital ECG format.
- Ten documented experimental stages, with 81 models trained to (nearly) 100 % training accuracy and evaluated on held-out patients.

- A best all-patient four-class accuracy of **0.558** from image-derived inputs only, and a triage accuracy of **0.696** on 48 % of patients.
- Training and held-out accuracy reported side by side for every model, with every results table sorted by training accuracy.

1.5 Organisation of the report

Chapter 2 reviews the literature. Chapter 3 describes the data, digitisation, features, networks and evaluation protocol. Chapter 4 reports the experimental study phase by phase and Chapter 5 the public-data transfer. Chapter 6 presents the results and their analysis, and Chapter 7 concludes with the plan for Phase II.

Chapter 2

Literature Review

2.1 Ejection fraction and its ECG signature

Clinical guidelines divide LVEF into normal, mildly reduced and reduced ranges [1, 2], and echocardiography is the standard way to measure it [5]. The measurement itself is not exact: two echocardiograms of the same patient may disagree by around 10 EF points [6], which is close to the width of a single class in this project (10–13 points). Several ECG findings are known to accompany a weak left ventricle: a prolonged QRS or bundle-branch block [7], a fragmented QRS [8], a wide angle between the QRS and T vectors [9] and uneven repolarisation [10, 11].

2.2 AI on the ECG for LV dysfunction

On digital 12-lead ECGs, a convolutional network flagged EF ≤ 35 % with an AUC of 0.93 [3], and in a randomised primary-care trial it led to more new diagnoses [4]. Fewer studies grade the EF instead of flagging it: Vaid et al. predicted three EF bands and also regressed the EF value, using 715,890 ECGs [12]. Working from ECG *images* rather than signals, a model trained on 385,601 ECGs reached AUROC 0.90–0.95 for EF < 40 % [13], and turning ECG images back into signals is itself an active research topic [14–16]. Models that predict several structural heart diseases together improved as more input types were added [17], and single-lead and wearable devices have carried the screening idea outside the hospital [18–20].

2.3 Foundation models and public data

Table 2.1 Relevant studies: data, input, task and main reported result (as printed in each abstract).

Study	Data	Input	Task	Main reported result
Attia et al. [3]	44,959 train / 52,870 test patients, Mayo Clinic	digital 12-lead	EF $\leq$ 35 %	AUC 0.93, accuracy 85.7 %
Attia et al. [32]	3,874 prospective patients	digital 12-lead	EF $\leq$ 35 %	AUC 0.918; 39.8 % of false positives had EF 36–50 %
Yao et al. [4]	22,641 adults, 120 care teams (RCT)	digital 12-lead	new low-EF diagnosis	diagnoses 1.6 % $\rightarrow$ 2.1 % (OR 1.32)
Adedinsewo et al. [33]	1,606 emergency patients	digital 12-lead	EF $\leq$ 35 % / $<$ 50 %	AUROC 0.89 / 0.85
Vaid et al. [12]	715,890 ECGs, 147,636 patients, 5 hospitals	digital 12-lead	EF 3 classes + regression	AUC 0.94 / 0.82 / 0.89 (internal); MAE 5.8–6.1
Kwon et al. [34]	55,163 ECGs, 2 hospitals	ECG features + demographics	HFrEF (EF $\leq$ 40 %)	AUROC 0.843 internal, 0.889 external
Sangha et al. [35]	2.2 million ECGs plotted as images	ECG images	6 rhythm / conduction labels	mean AUROC 0.99 (external 0.97)
Sangha et al. [13]	385,601 ECGs, Yale + external sites	ECG images	EF $<$ 40 %	AUROC 0.91 internal, 0.90–0.95 external
Ulloa-Cerna et al. [17]	2.2 million ECGs, 484,765 adults	ECG + age, sex	composite structural disease	AUROC 0.91
Elias et al. [36]	77,163 patients	digital 12-lead	moderate / severe AS, AR, MR	AUROC 0.88 / 0.77 / 0.83
Poterucha et al. [23]	$>$ 1 million ECG–imaging pairs	digital 12-lead	structural heart disease	outperformed cardiologists; data and weights released
Hughes et al. [24]	100,000 ECGs, 36,286 patients (public)	digital, 250 Hz	composite + 11 labels	composite AUROC 82.0
McKeen et al. [22]	1.5 million ECGs (pre-training)	digital 12-lead	EF $\leq$ 40 % and others	AUROC 0.929 for EF $\leq$ 40 %
Li et al. [21]	10.8 million ECGs, 150 labels	digital 12-lead	broad diagnosis	AUROC $>$ 0.95 for 80 diagnoses

Table 2.1 (continued)

Study	Data	Input	Task	Main reported result
This work	837 development printouts, one centre; EchoNext 4,827	printout images (and derived traces)	**four-class EF**	0.558 all patients; 0.696 (48 % answered); heart-failure screen AUC up to 0.835

Foundation models change what a small dataset can achieve. ECGFounder learnt from 10.8 million ECGs and adapts to new tasks with few labelled examples [21], and ECG-FM reported AUROC 0.929 for EF $\leq$ 40 % [22]. The EchoNext project made paired ECG–echocardiography data and trained models public, and its models outperformed cardiologists [23, 24]. Earlier one-dimensional networks had already matched cardiologists in rhythm classification [25] and automated 12-lead interpretation [26]. For images, this project uses the ResNet [27], EfficientNet [28], MobileNetV3 [29] and ConvNeXt [30] backbones; for sequences, the LSTM [31] and convolutional LSTM variants.

2.4 Research gaps addressed in Phase I

- **Graded EF from printouts.** Most published models answer a yes/no question on digital signals. Four-class grading that includes Mild, from printout images, had not been reported, and Mild in particular needs more data than single-centre studies usually have.
- **Small single-centre data.** Methods for a few hundred labelled printouts, evaluated with nested cross-validation, are rarely studied.
- **Transfer to the printout format.** Public ECG–echo data had not been redrawn in a hospital's printout format for pre-training.
- **Confident subsets.** Triage that answers only confident cases and refers the rest is seldom evaluated for graded EF.

Chapter 3

Theoretical Study and Methodology

3.1 System overview

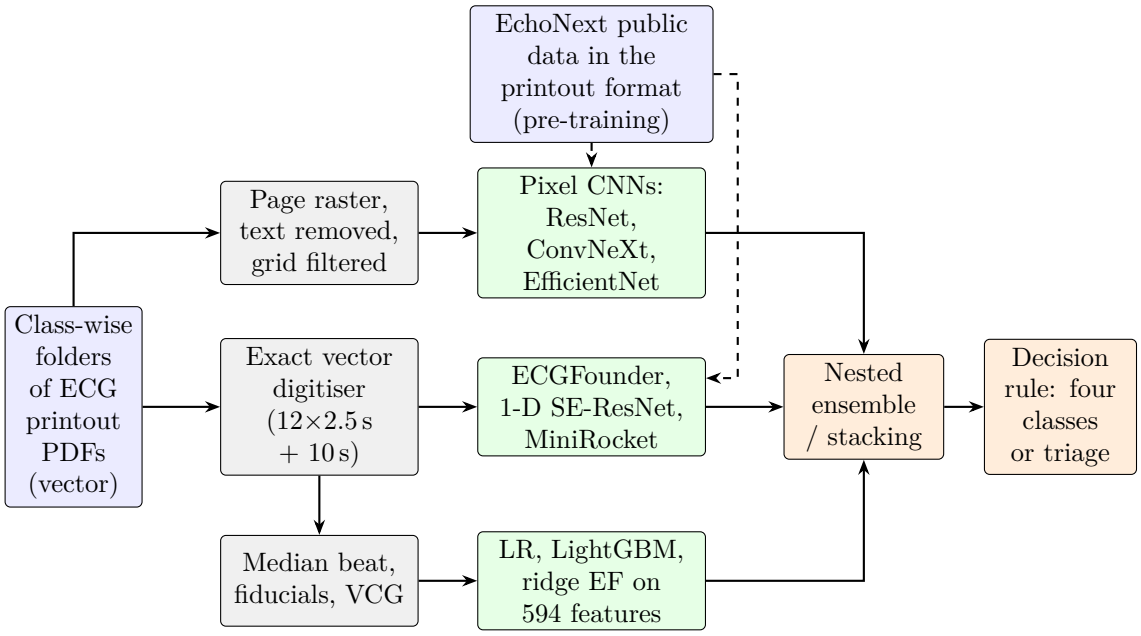


Figure 3.1 System architecture of the Phase I pipeline. Dashed arrows: knowledge transferred from the EchoNext public dataset.

The Phase I pipeline (Figure 3.1) has three parts. The printout PDF is digitised exactly, turned into image, signal and feature representations, and passed to the model families. External knowledge enters from the public EchoNext data, redrawn in the hospital printout format.

3.2 Dataset

3.2.1 ECG and echocardiography data collection

The data come from one tertiary-care hospital and cover 2020–2024: 1,047 ECG printouts of 993 patients, each with an echocardiography report. Every ECG is a standard 12-lead recording stored as a PDF, printed in the usual 3×4 layout (2.5 s per lead) with a 10-s lead II rhythm strip, at 25 mm/s and 10 mm/mV. The LVEF in the echocardiography report is the reference value from which each ECG's class is derived. Name, age, sex and examination date were used to link the ECG and echocardiography records.

3.2.2 Spreadsheet loading and cleaning

The ECG and echocardiography details were kept in separate Excel files, which were read into Python with the Pandas library. Before matching, patient names were normalised (extra spaces, special characters and different name orders removed or unified) so that the same person is written the same way in both files. Only the fields needed later were kept: patient name, age, sex, examination date, ECG file path and LVEF value. Records with missing or unusable entries were flagged so that they could not cause false matches.

3.2.3 Patient record matching with fuzzy matching

Because the same name was not always spelt identically in the two files, exact matching would have missed many pairs. Names were therefore compared with fuzzy string matching (RapidFuzz library), which gives a similarity score and tolerates small spelling differences, swapped word order and spacing. Every candidate pair was then checked against age, sex and examination date, which removed most wrong matches and made the paired dataset more reliable.

3.2.4 LVEF extraction and class labels

For every matched pair the EF value was read from the echocardiography record and converted to a number. Each ECG then received one of the four classes defined in

Section 1.2 (Normal, Mild, Moderate, Severe), and this label was stored with the ECG record for all later steps.

3.2.5 Class-wise organisation of the ECG files

Using the file paths in the matched table, the ECG files were copied into four folders, one per class (Normal, Mild, Moderate, Severe). This structure makes every class easy to access during preprocessing and training.

3.2.6 Dataset partitioning

To keep the evaluation fair, the 1,047 ECGs were split once *by patient*, so that no patient appears in both parts. The **development set** holds 837 ECGs (384 Normal, 185 Mild, 177 Moderate, 91 Severe) and was used for all modelling. The **locked test set** holds the other 210 ECGs (96 / 47 / 44 / 23); it was kept apart during development and is not used for any result in this report.

3.3 Exact digitisation and the printout renderer

The PDFs are vector drawings: each lead is a polyline of about 700 segments, so the trace is recovered exactly. With the row baselines y_{base} and column origins x_0 of the page,

$$v(t) = \frac{y_{\text{base}} - y(x)}{82}\,\text{mV}, \qquad t = \frac{x - x_0}{205.2}\,\text{s}, \tag{3.1}$$

resampled to 500 Hz. The inverse operation, a *printout renderer*, draws any 12-lead signal on a page identical to the hospital's, rounding each point to whole drawing units as the machine does. Redrawn development ECGs match their real pages: 81 % of drawn points are identical and the read-back error is 5–13 μ V.

3.4 Input representations and features

Four representations were built from each printout:

- **page image**: the waveform area with text removed and the grid filtered, for CNNs;

- **aligned signal**: the 12 leads in their own 2.5-s windows on a common 10-s axis, for ECGFounder and the 1-D networks;
- **ECGFounder embeddings**: the 1,024-d final embedding and 150 diagnostic outputs;
- **engineered features**: 594 features, among them 191 clinician-style and vectorcardiographic measurements such as the spatial QRS–T angle, total cosine R-to-T (TCRT) and the QT variability index.

3.5 Network architectures, preprocessing filters and training

3.5.1 Preprocessing filters

Each representation has its own filter chain (Table 3.1): a colour filter removes the pink grid from page images, and a 50 Hz notch, a 0.67–40 Hz band-pass and a moving-median baseline filter clean the signals.

3.5.2 Network layers and activation functions

Table 3.2 gives the layers, kernels, activation functions, normalisation and size of every network. The activation functions used are

$$\begin{gathered}\mathrm{ReLU}(x)=\max(0,x), \qquad \mathrm{SiLU}(x)=x\,\sigma(x), \qquad \mathrm{GELU}(x)=x\,\Phi(x),\\ \text{h-swish}(x)=x\,\frac{\mathrm{ReLU6}(x+3)}{6}, \qquad \sigma(x)=\frac{1}{1+e^{-x}},\\ \mathrm{softmax}(\mathbf{z})_c=\frac{e^{z_c}}{\sum_{j=1}^{4} e^{z_j}},\end{gathered} \tag{3.2}$$

where Φ is the standard normal distribution function. Every four-class network ends in a softmax, and ordinal heads use sigmoids for $P(y>k)$.

3.5.3 Training settings and training accuracy

Every model in this report was trained until its training accuracy reached (nearly) 100 %, for as many epochs or iterations as it needed; the settings of every model family are

Table 3.1 Preprocessing filters and their parameters for each input representation.

Input	Step	Filter / operation and parameters
Page image (CNNs)	rasterise and crop	vector PDF rendered at 1 px/pt; crop to the waveform area (removes all header text)
	grid removal	colour filter: darkness $= 255 - \max(R, G, B)$ (HSV value); the pink grid has a high value and vanishes
	binarisation	Otsu threshold (clamped)
	denoising	connected components < 6 px removed (keeps 1-px trace lines that a median filter would erode)
	thickening, resizing	scale-aware dilation; area interpolation to 512×1024, then 256×512 network input
	filter ablations	Gaussian 5×5, $\sigma = 1$; median 3×3; bilateral (d 7, σ 50/7); CLAHE (clip 2.0, 8×8 tiles); adaptive Gaussian threshold (block 25, C 60)
Digitised signal (ECGFounder)	exact digitisation	vector paths converted at 82 pt/mV and 616/3 pt/s; 500 Hz
	power-line notch	IIR notch at 50 Hz, $Q = 30$
	band-pass	4th-order Butterworth, 0.67–40 Hz, zerophase (forward–backward)
	baseline	0.4-s moving-median baseline subtracted
	scaling, layout	z-score per lead; 12 leads in their own 2.5-s windows on a 10-s axis ($12 \times 5{,}000$)
Engineered features	beat analysis	0.5 Hz high-pass; R-peak detection; Raligned median beats; vectorcardiogram by the Kors matrix
EchoNext signals	harmonisation	de-normalisation to μ V; resampling $250 \rightarrow 500$ Hz; cascaded median filters (0.2 s, 0.6 s) for baseline removal; clipping; printout cut (2.5-s windows)
Pixel CNNs (IMG)	page image	waveform area, text removed, grid filtered; 320×640 (ResNet-18) or 384×768 (ConvNeXt-Tiny, EfficientNet-B2); grey channel repeated to RGB

listed in Table 3.6. Image CNNs and the 1-D SE-ResNet were trained with AdamW (learning rate 10^{-3} after a short warm-up, then constant), batch 16 and class-weighted cross-entropy, without augmentation, label smoothing, weight averaging, weight decay or early stopping, until their training accuracy reached at least 99.5 %; no network needed its epoch cap. ECGFounder was fine-tuned in the same way with differential learning rates (backbone 10^{-4}, head 10^{-3}). The MLP heads were trained without weight decay or early stopping until they reached 100 % (at most 3,000 epochs), the linear and logistic models with almost no regularisation (C = 10,000 or L2 10^{-6}), and gradient boosting with 2,000 trees, one sample per leaf allowed and no shrinkage

penalty. Table 3.3 compares training and validation accuracy, and Table 3.5 shows how the twelve main models are built.

- **Image and signal networks** reach 0.996–1.000 training accuracy in 12–57 epochs, but 0.35–0.56 on the 168 validation ECGs. Their training loss falls below 0.04 while the validation loss rises to 1.7–6.4 (Figure 3.2): they memorise the training ECGs, and every CNN reached its highest validation accuracy before its last epoch.
- **Foundation-feature heads** reach 1.000 on the training folds against 0.49–0.54 held out. ECGFounder fine-tuning reaches 0.996 in 12 epochs and gives the best validation accuracy of the fixed split (0.560). A logistic regression on header and clinical features stops at 0.589 even without regularisation, because straight-line class boundaries cannot separate every training ECG.
- With 837 ECGs for 30.7 million ECGFounder parameters, frozen features, nested selection and ensembling were used in all later phases.

3.5.4 Analysis of transfer learning

Transfer learning was used in four forms, from natural images to EF-labelled ECGs (Table 3.4). The closer the source is to the target, the larger the gain. On the same fixed split, ECG pre-training (ECGFounder fine-tuned, 0.560) beat ImageNet pre-training (ResNet-18, 0.423) by 13.7 points, and EF-labelled public ECGs (EchoNext) supplied the members behind the best all-patient ensembles: the best ensemble reaches 0.558 with its EchoNext members and 0.539 without them. Models trained only on US patients reach up to 0.539 four-class accuracy on our printouts with only their decision thresholds tuned on hospital data, which also validates the printout conversion, and a pixel CNN trained only on EchoNext ECGs drawn as hospital printouts reaches 0.505 on the hospital pages.

3.6 Evaluation protocol

- **Cross-validation:** patient-grouped 5-fold CV repeated three times (15 cells), identical folds for every model.

Table 3.2 Layers, activation functions, normalisation and size of every network in this report.

Network	Layers and blocks	Activation	Norm.	Params
ResNet-18	7×7 conv stem + max-pool; 4 stages $\times$ 2 basic blocks of two 3×3 convs (64–512 channels); global average pool; FC(4) — 18 weight layers	ReLU	BatchNorm	11.2 M
ECGNet (custom)	$7 \times 7/2$ stem; 4 stages of 1–1–2–2 residual SE blocks (widths 64–256) with time-elongated 3×5 kernels; [GAP ‖ GMP]; dropout 0.3; FC(4) —13 conv layers	ReLU; sigmoid (SE)	BatchNorm	6.3 M
EfficientNet-B0	3×3 stem; 16 MBConv blocks in 7 stages with SE; 1×1 head conv; FC(4)	SiLU	BatchNorm	4.0 M
MobileNetV3-Large	3×3 stem; 15 inverted-residual bottlenecks, SE in 8; 1×1 head	h-swish, ReLU	BatchNorm	4.2 M
ConvNeXt-Atto	4×4 patchify stem; 4 stages of 2–2–6–2 blocks (7×7 depth-wise conv + MLP)	GELU	LayerNorm	3.4 M
1-D SE-ResNet	1-D residual blocks, kernel 9, SE; [GAP ‖ GMP]; FC(4)	ReLU	BatchNorm	4.5 M
ECGFounder (Net1D)	1-D conv stem (kernel 16, stride 2); 7 stages of 2–2–2–3–3–4–4 blocks (20 blocks; 64–1,024 filters), grouped kernel-16 convs with SE; GAP $\rightarrow$ 1,024-d embedding	Swish	none	30.7 M
MLP head / fusion	dropout 0.5 $\rightarrow$ linear 64 $\rightarrow$ GELU $\rightarrow$ dropout 0.5 $\rightarrow$ linear 4	GELU	standardised input	66 k
Meta-learners	multinomial logistic regression; Frank–Hall ordinal (3 binary logistic models)	softmax; sigmoid	standardised	< 1 k

- **Nested choices:** every hyper-parameter, class weight, threshold and ensemble member is chosen inside the training folds [37]. Class weights maximise accuracy subject to every recall ≥ 0.18 (fallback 0.15):

$$\hat{y} = \arg\max_k \, w_k P_k, \qquad \mathbf{w}^* = \arg\max_{\mathbf{w}} \; \mathrm{Acc}(\mathbf{w}) \;\; \text{s.t.} \;\; \min_k \mathrm{Rec}_k(\mathbf{w}) \geq 0.18. \tag{3.3}$$

- **Intervals and tests:** corrected resampled t statistic [38] with Holm adjustment [39]:

$$t = \bar{d} \Big/ \sqrt{\Big(\frac{1}{J} + \frac{n_2}{n_1}\Big)\hat{\sigma}_d^2}, \qquad J = 15. \tag{3.4}$$

- **Integrity:** every study, including the training to 100 %, was pre-registered in a dated document before it was run; the locked test set is not used in this report.

Table 3.3 Training against validation accuracy of the main networks, every network trained until its training accuracy reached (nearly) 100 % (no early stopping, augmentation, weight decay, label smoothing or weight averaging). Training accuracy is measured on all training ECGs in evaluation mode. Sorted by training accuracy within each part. Epochs: until the target was reached (CNNs ≥ 99.5 %); solver / iter. = models not trained in epochs.

Network	Epochs	Training accuracy	Validation / held-out accuracy	Train / val loss
Networks of the initial project and Phase 1, fixed split (669 training / 168 validation ECGs)				
EfficientNet-B0	37	1.000	0.417	0.002 / 5.230
1-D SE-ResNet	32	0.999	0.435	0.014 / 3.005
ECGNet	21	0.999	0.435	0.005 / 3.354
MobileNetV3-Large	57	0.999	0.387	0.003 / 5.845
ResNet-18	24	0.997	0.423	0.010 / 2.702
ECGFounder fine-tuned (stages 5–6)	12	0.996	0.560	0.036 / 1.657
ConvNeXt-Atto	53	0.996	0.345	0.025 / 6.426
Frozen-feature heads, development cross-validation (mean of the 15 folds)				
FUSION-MLP (header + ECGFounder, MLP head)	112	1.000	0.538	–
ECGF-MLP (ECGFounder + MLP head)	108	1.000	0.536	–
ECGF-linear (ECGFounder + linear head)	100 iter.	1.000	0.491	–
HDRCLIN-LR (header + clinical features, logistic regression)	solver	0.589	0.486	–

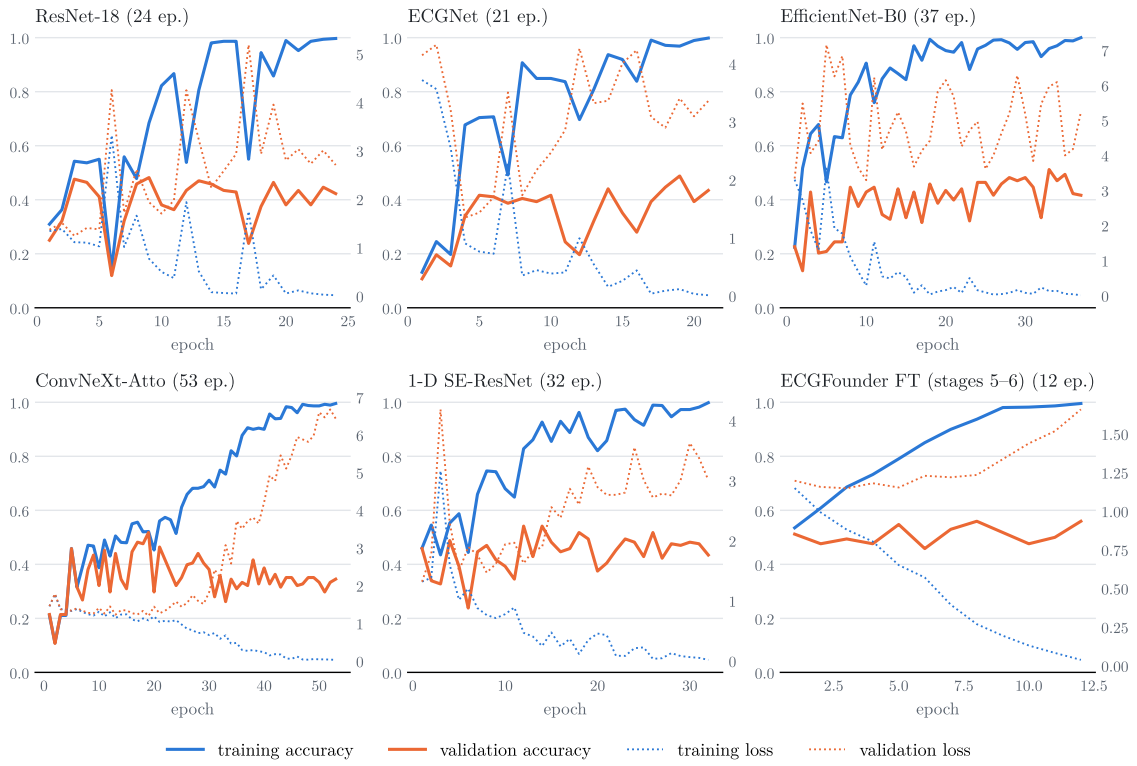


Figure 3.2 Training curves of six representative networks trained until their training accuracy reached ≥ 99.5 %: training and validation accuracy (solid) and loss (dotted) per epoch, fixed split.

Table 3.4 Transfer learning: source, how the knowledge was transferred and what it gave on the hospital printouts (every model trained to (nearly) 100 % training accuracy).

Source	**Transfer method**	**Result on the hospital data**
ImageNet (1.3 M natural images)	initialise ResNet-18; train on the hospital page images	0.997 training, 0.423 validation (fixed split; best 0.482 before the last epoch): natural-image features transfer only partly to thin line drawings
ECGFounder (10.8 M ECGs, 150 labels)	frozen 1,024-d embedding + linear or MLP head	1.000 training, 0.491–0.536 held out (CV); EF regression r 0.624; the backbone of every main model
	partial fine-tuning (stages 5–6)	0.996 training, 0.560 validation (fixed split): the best network on that split
EchoNext signals (4,827 US ECG–echo pairs)	models trained only on EchoNext; only their decision thresholds tuned on hospital data	0.78–1.00 training and 0.46–0.76 validation accuracy on EchoNext; 0.41–0.54 four-class accuracy on the hospital ECGs
	EchoNext models as ensemble members	SUPER ensembles 0.548–0.558, the best all-patient results; 0.539 for the same recipe without them
EchoNext images (drawn as printouts)	pixel CNN trained on the printout images	0.999 training, 0.725 validation on EchoNext; 0.505 on the hospital pages

Table 3.5 How the twelve main models are built, sorted by training accuracy. ECGFounder (Net1D; 61 convolution and 40 linear layers in 7 stages of 20 residual blocks; 30.7 M parameters) was pre-trained on 10.8 million ECGs and is kept frozen, so only the heads are trained on the hospital ECGs. Epochs: median over the 15 folds to reach 100 % training accuracy; [b] epochs of the member FUSION-MLP; [c] epochs of the member ECGF-MLP.

Model	Inputs	Layers / structure	Params	Epochs	Training acc.
Triage†	ECGFounder + 118 clinician features → EF; IMG-C3	ridge + LightGBM EF regressors → EF grade (3 cut-points) + margin; answers when the grade agrees with IMG-C3	regressors + 4	4,000 trees	1.000
LGBM-FEAT	16 header + 403 engineered waveform features (419)	2,000 boosting rounds × 4 classes, 31 leaves per tree	8,000 trees	2,000 trees	1.000
FUSION-MLP	16 header + ECGFounder embedding (1,040)	MLP head as ECGF-MLP; 3 seeds	67.0 k trained	112	1.000
ECGF-MLP	frozen ECGFounder embedding (1,024)	frozen backbone + MLP: dropout, linear 1,024 → 64, GELU, dropout, linear 64 → 4; 3 seeds	65.9 k trained	108	1.000
IMG-C3	IMG-vote	IMG-vote + nested prior rule	members + 1	108[c]	1.000
C1	FUSION-MLP + ECGF-linear + HDRCLIN-LR	average of the 3 members' probabilities	members	112[b]	1.000
IMG-vote	ECGF-MLP + ECGF-linear (image only)	average of the 2 members' probabilities	members	108[c]	1.000
ECGF-linear	frozen ECGFounder embedding (1,024)	frozen backbone + 1 linear layer (1,024 → 4)	4.1 k trained	100 iter.	1.000
C3	C1	C1 + nested prior rule (one α chosen on inner out-of-fold predictions)	members + 1	112[b]	0.995
CLIN-LR	118 image-only clinician features	1 linear layer (118 → 4)	476	solver	0.720
HDRCLIN-LR	16 header + 28 clinical ECG features (44)	1 linear layer (44 → 4)	180	solver	0.589
HDR-LR	16 header measurements (age, sex, rate, intervals, axes)	1 linear layer (16 → 4)	68	solver	0.529

Table 3.6 Training settings of every model family. Inputs, labels, architectures, folds and decision rules are those of each stage; every model was trained for as many epochs or iterations as it needed, until its training accuracy reached (nearly) 100 %.

Model family	**Training**
MLP heads and fusion networks	AdamW, weight decay 0, no early stopping; until 100 % training accuracy (at most 3,000 epochs)
Linear and logistic models	almost no regularisation (L2 10^{-6} or C = 10,000), solved to convergence
Gradient boosting	2,000 trees (4,000 for the EF regressor), one sample per leaf allowed, L2 0, no early stopping
Image CNNs, 1-D SE-ResNet, ECGFounder fine-tuning	AdamW, constant learning rate after a short warm-up; no augmentation, label smoothing, weight averaging or weight decay; until training accuracy $\geq$ 99.5 % (epoch caps 100 / 150 / 60, never reached)
k-nearest neighbours	$k = 15$, distance weighting
Ensembles, stacks, decision rules	built from the members above; weights, thresholds and prior rules chosen on inner out-of-fold predictions; meta-learners C = 10,000

Chapter 4

Experimental Study

Table 4.1 Overview of the experimental stages of Phase I, with the main outcome of each stage after every model was trained to (nearly) 100 % training accuracy.

Stage	**Main question**	**Evaluation**	**Outcome carried forward**
Initial project	Can a page-image CNN grade LVEF?	fixed split (168 validation ECGs)	10 models; CNNs 0.345–0.435, XGBoost 0.494
Phase 1	Does a foundation ECG model help?	fixed split	ECGFounder fine-tuned 0.560, the best network
Phase 2	Robust re-evaluation, pre-registered	development CV	LGBM-FEAT 0.548; FUSION-MLP macro-F1 0.438
Phase 3	Fusion and ensembles	development CV	vote ensemble C1 0.517
Phase 4	EF regression, multitask	development CV	EF $r = 0.624$; multitask 0.536
Phase 5	Features, boosting, decision rules	development CV	E2 LightGBM 0.557; C3 selected, 0.533
Phase 6	Fine-tuning the foundation model	fixed split	ECGFounder fine-tuned 0.560
Label studies	Cleaning and cohorts	CV and subsets	0.67–0.69 on the clean subsets
Phase 7	Representations and triage	development CV	triage 0.696 on 48 % of patients
Image-only push	HIER, ENH, ULT, EchoNext, SUPER	development CV	ENH-vote 0.549; SUPER-TOP10-ORD 0.558

Phase I was carried out in ten stages (Table 4.1). Each stage asked one question, and its lesson was carried into the next. Every model was trained, with the inputs, labels, folds and decision rules of its stage, until its training accuracy reached (nearly)

100 % (Section 3.5.3); the results below are those of 79 classifiers and two EF regressors. Every table is sorted by training accuracy, highest first, and models with the same training accuracy by held-out accuracy. Figure 6.2 in Chapter 6 compares all of them.

4.1 Initial project and Phase 1

Table 4.2 Initial project and Phase 1: image and signal networks on the fixed split (trained on 669 ECGs, validated on 168), every network trained until its training accuracy reached (nearly) 100 %; sorted by training accuracy, ties by validation accuracy. Best valid. = highest validation accuracy at any epoch on the way to 100 %. On 168 ECGs the 95 % interval of an accuracy is about ±0.075.

Model	Layers	Params	Epochs	Training acc.	Validation acc.	Best valid.
XGBoost (waveform features)	2,000 trees on 263 waveform features	–	–	1.000	0.494	–
HOG + RBF-SVM	HOG + PCA-150 + RBF kernel	–	–	1.000	0.423	–
EfficientNet-B0	81 conv (16 MBConv) + 1 FC	4.0 M	37	1.000	0.417	0.512
1-D SE-ResNet	37 conv + 1 FC; digitised waveforms	4.5 M	32	0.999	0.435	0.542
ECGNet	29 conv (incl. SE) + 1 FC	6.3 M	21	0.999	0.435	0.488
MobileNetV3-Large	63 conv (15 bottlenecks) + 1 FC	4.2 M	57	0.999	0.387	0.476
ResNet-18	20 conv + 1 FC (18 weight layers); ImageNet start	11.2 M	24	0.997	0.423	0.482
ECGFounder fine-tuned (stages 5–6)	61 conv + 40 linear (stages 5–6 trained) + 1 FC	30.7 M	12	0.996	0.560	0.560
ConvNeXt-Atto	40 conv (12 blocks) + 1 FC	3.4 M	53	0.996	0.345	0.518
Header 0.6 + ResNet-18 0.4	weighted average (default pipeline of the initial project)	–	–	0.972	0.458	–
Logistic regression (header)	1 linear layer on the header	–	–	0.674	0.405	–

The initial project compared image and signal models on a fixed split of the development data: classical HOG features with an SVM, CNNs trained from scratch (ECGNet, EfficientNet-B0, MobileNetV3, ConvNeXt-Atto), ImageNet ResNet-18, a 1-D SE-ResNet on digitised waveforms, gradient boosting (XGBoost) on 263 engi-

neered features and the printed header (Table 4.2). Trained to (nearly) 100 %, every network fits its 669 training ECGs (0.996–1.000), but on the 168 validation ECGs the CNNs reach only 0.345–0.435 and the default header + ResNet-18 pipeline 0.458. Every CNN reached its highest validation accuracy (0.476–0.542) before its last epoch. Phase 1 introduced ECGFounder: fine-tuned in stages 5–6, it reaches 0.560 after 12 epochs, the best result on this split, and it became the backbone of later work.

4.2 Phases 2–5: robust evaluation, fusion and decision rules

Table 4.3 Phases 2–3 and 5: the twelve main models in development cross-validation (5 folds × 3 repeats, 837 ECGs), each trained to (nearly) 100 % training accuracy; sorted by training accuracy, ties by held-out accuracy. Held-out accuracy: mean of the 15 test folds with the corrected 95 % CI; F1, balanced accuracy and recall from the confusion matrix pooled within each repeat. HF AUC = heart-failure screen, reduced EF against Normal (C3 and IMG-C3 use the probabilities of C1 and IMG-vote). † Selective: scored on the patients it answers (share in brackets).

Model	Training acc.	Held-out acc. [95 % CI]	Macro F1	Mild F1	Bal. acc.	HF AUC	Recall N / Mi / Mo / S
Image-only triage†	1.000 (58 %)	0.696 [0.630, 0.762] (48 %)	0.372	0.049	0.367	–	0.99 / 0.03 / 0.19 / 0.27
LGBM-FEAT	1.000	0.548 [0.512, 0.583]	0.409	0.227	0.413	0.824	0.88 / 0.18 / 0.39 / 0.21
FUSION-MLP	1.000	0.538 [0.499, 0.578]	0.438	0.255	0.440	0.831	0.79 / 0.24 / 0.42 / 0.31
ECGF-MLP	1.000	0.536 [0.506, 0.565]	0.430	0.282	0.430	0.835	0.79 / 0.27 / 0.41 / 0.26
IMG-C3	1.000	0.532 [0.492, 0.572]	0.416	0.252	0.413	0.822	0.82 / 0.23 / 0.38 / 0.22
C1	1.000	0.517 [0.477, 0.558]	0.426	0.265	0.426	0.824	0.76 / 0.26 / 0.38 / 0.32
IMG-vote	1.000	0.503 [0.463, 0.544]	0.416	0.275	0.415	0.822	0.72 / 0.28 / 0.39 / 0.27
ECGF-linear	1.000	0.491 [0.451, 0.532]	0.408	0.282	0.407	0.777	0.70 / 0.29 / 0.36 / 0.28
C3	0.995	0.533 [0.496, 0.569]	0.423	0.260	0.419	0.824	0.81 / 0.24 / 0.39 / 0.24
CLIN-LR	0.720	0.442 [0.390, 0.493]	0.376	0.240	0.382	0.767	0.60 / 0.25 / 0.36 / 0.32
HDRCLIN-LR	0.589	0.486 [0.436, 0.536]	0.415	0.252	0.429	0.786	0.68 / 0.24 / 0.35 / 0.45
HDR-LR	0.529	0.483 [0.432, 0.535]	0.391	0.210	0.409	0.785	0.72 / 0.17 / 0.32 / 0.42

Phase 2 moved all modelling to pre-registered, patient-grouped cross-validation on the 837 development ECGs (Table 4.3, Figure 4.1). Nine of the twelve main models reach 1.000 training accuracy and C3 0.995; the three logistic regressions stop at 0.53–0.72 even without regularisation. On held-out patients LightGBM on header and engineered features is the most accurate main model (0.548), FUSION-MLP has the highest macro-F1 (0.438) and balanced accuracy (0.440), and the ECGFounder heads find the Mild class best (Mild F1 0.282). For heart-failure screening (reduced EF against Normal) the ECGFounder-based models reach AUC 0.78–0.84 (ECGF-MLP

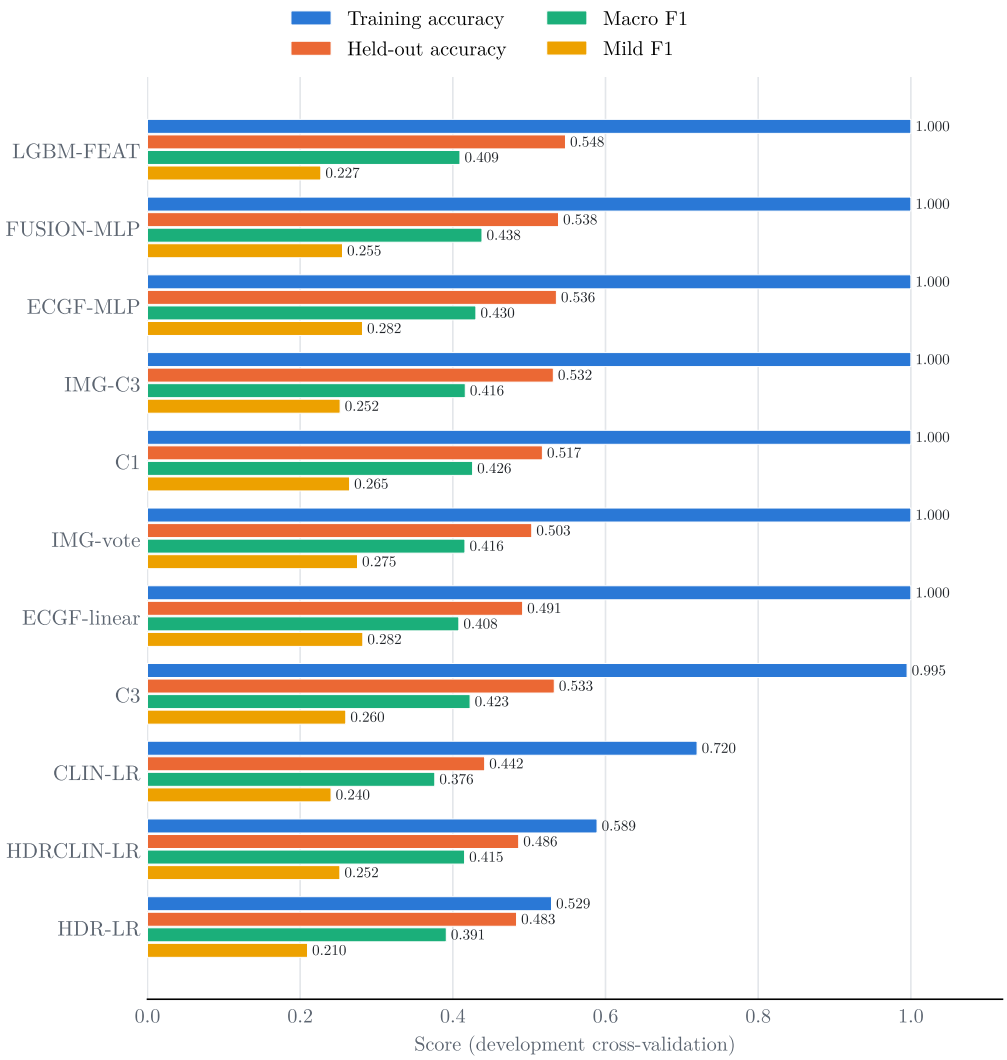


Figure 4.1 Training accuracy, held-out accuracy, macro-F1 and Mild F1 of the eleven four-class main models after training to (nearly) 100 % training accuracy, sorted by training accuracy (development cross-validation).

0.835) and the header and clinical logistic regressions 0.77–0.79. Phase 3 combined representations: the soft-vote ensemble C1 reaches 0.517.

Phase 5 tested engineered features, boosting and decision rules (Table 4.4). E2, LightGBM on the engineered features with a 32-component projection of the ECG-Founder embedding, is the most accurate (0.557 [0.517, 0.597]). The selected model C3, a vote ensemble with a nested class-prior rule, reaches 0.533 with every class detected.

Table 4.4 Phase 5 experiments in development cross-validation, every model trained to (nearly) 100 % training accuracy; sorted by training accuracy. Min recall = lowest recall of the four classes (admissible if ≥ 0.15). Epochs: median over the 15 folds; [b] epochs of the member FUSION-MLP.

Experiment	Epochs	Training acc.	Held-out acc. [95 % CI]	Min recall	Macro F1	Note
E2 LightGBM, features + ECGFounder PCA	2,000 trees	1.000	0.557 [0.517, 0.597]	0.162	0.422	admissible
E1 MLP, header + ECGFounder + feature families	64	1.000	0.542 [0.504, 0.579]	0.238	0.442	admissible
E7 mixup MLP	144	1.000	0.534 [0.499, 0.570]	0.243	0.434	admissible
E4 hierarchical MLP	52 / 39	0.999	0.540 [0.517, 0.563]	0.223	0.433	admissible
E5 vote + nested prior rule (C3)	112[b]	0.995	0.533 [0.496, 0.569]	0.238	0.423	admissible
E6 signal-quality gating	112[b]	0.988	0.542 [0.500, 0.583]	0.232	0.441	admissible
E3 weighted average of the C1 members	112[b]	0.985	0.530 [0.480, 0.581]	0.223	0.428	admissible
E3 stacking (logistic meta-learner)	112[b]	0.826	0.532 [0.487, 0.576]	0.132	0.418	a recall below 0.15

4.3 Phases 4 and 6: EF regression, multitask learning and fine-tuning

Table 4.5 Phases 4 and 6: EF regression, multitask learning and fine-tuning, every model trained to (nearly) 100 % training accuracy (EF regressors: training correlation ≈ 1); sorted by the training score. Development cross-validation unless stated.

Model	Measure	Epochs	Training	Held-out	Note
Multitask MLP (four classes + EF, header + ECGFounder)	four-class accuracy	131	1.000	0.536	held-out 95 % CI [0.496, 0.577]
EF regression, ridge + LightGBM (triage regressor)	EF (correlation r)	4,000 trees	1.000	0.476	MAE 0.0 / 13.8 EF points
ECGFounder fine-tuned (stages 5–6), fixed split	four-class accuracy	12	0.996	0.560	168 validation ECGs ($\pm$0.075)
EF regression MLP (header + ECGFounder)	EF (correlation r)	3,000 (cap)	0.993	0.624	MAE 2.7 / 11.0 EF points

Phase 4 added EF regression from the image (Table 4.5). The MLP regressor reproduces the training EF almost exactly ($r = 0.993$, mean error 2.7 points) but reaches $r = 0.624$ and a mean error of 11.0 points on held-out patients; the ridge +

LightGBM regressor used by the triage model fits its training EF exactly and reaches $r = 0.476$. Learning the class and the EF together (multitask) gives 0.536. Phase 6 fine-tuned ECGFounder, which gives the best fixed-split result (0.560, Table 4.2).

4.4 Label-quality and cohort studies

Table 4.6 Label-quality and cohort studies with the C3 recipe trained to (nearly) 100 % training accuracy (development cross-validation); sorted by training accuracy. Accuracy on the subset named, with the recorded labels. Epochs: median of the member FUSION-MLP.

Study	Scored on	Epochs	Training acc.	Accuracy	Finding
NF: noise-free training (all classes)	264 clean, clear-cut ECGs	45[b]	1.000	0.673	clean labels are graded well
HC30: train on exact-name links within 30 days	395 reliable links	52[b]	1.000	0.583	reliable links give about 0.58
CLEAR: train without the 255 border ECGs	582 clear-cut ECGs	104[b]	0.993	0.670	clear cases are graded well
VER: 15 labels corrected to a high-confidence echo	all 837	132[b]	0.986	0.537	label noise costs little
NFS: noise-free training, Severe kept	264 clean, clear-cut ECGs	44[b]	0.985	0.694	best on clean labels
MMCLEAN: drop 217 suspect Mild or Moderate labels	620 kept ECGs	95[b]	0.978	0.669	accuracy follows label clarity

Pre-registered studies asked whether cleaner labels or reliable cohorts lift performance (Table 4.6). On ECGs with clear or verified labels the C3 recipe scores in the upper 60s: 0.670 on 582 clear-cut ECGs, 0.669 on the 620 ECGs kept after removing suspect Mild and Moderate labels, and 0.694 on 264 clean, clear-cut ECGs. Trained on reliable links only it reaches 0.583, and correcting 15 labels gives 0.537 on all ECGs. Clinician-style features identified interpretable Mild markers: a spatial QRS–T angle above 100° in 52 % of Mild against 20 % of Normal ECGs, and poor R-wave progression in 31 % against 7 %.

4.5 Phase 7 and the image-only models

Phase 7 built header-free cascades. The triage model AGT-img answers only when its EF grade agrees with the image-only classifier IMG-C3 and the EF estimate is far from every class limit: it is correct for **0.696 of the 48 %** of ECGs it answers

Table 4.7 Image-only four-class models (no printed header; all 837 ECGs, development cross-validation), every model trained to (nearly) 100 % training accuracy; sorted by training accuracy. [c] Epochs of the member ECGF-MLP; solver = fitted to convergence.

Model	Epochs	Training acc.	Held-out acc. [95 % CI]	Recall N / Mi / Mo / S	Macro F1
ENH-vote (6 members, tuned class weights)	108[c]	1.000	0.549 [0.513, 0.585]	0.87 / 0.18 / 0.40 / 0.24	0.418
ULT M9 k-nearest neighbours ($k = 15$)	none (kNN)	1.000	0.546 [0.520, 0.571]	0.90 / 0.13 / 0.43 / 0.11	0.382
ULT M10 gradient boosting	2,000 iter.	1.000	0.540 [0.496, 0.585]	0.86 / 0.19 / 0.39 / 0.21	0.409
HIER two-stage (Normal / middle / Severe, then Mild vs Moderate)	108[c]	1.000	0.535 [0.488, 0.582]	0.82 / 0.18 / 0.42 / 0.30	0.426
ENH M5 logistic regression on the HIER inputs	solver	1.000	0.501 [0.450, 0.553]	0.71 / 0.29 / 0.38 / 0.26	0.417
ENH M2 synthetic 10-s ECG → ECGFounder	solver	1.000	0.489 [0.439, 0.539]	0.69 / 0.27 / 0.40 / 0.26	0.404
ENH M4 MiniRocket	solver	1.000	0.469 [0.423, 0.515]	0.67 / 0.22 / 0.36 / 0.34	0.390
ENH M3 median-beat shape model	solver	1.000	0.412 [0.358, 0.466]	0.59 / 0.23 / 0.30 / 0.27	0.345
ULT M8 logistic regression, multi-scale ECGFounder	solver	1.000	0.496 [0.459, 0.532]	0.75 / 0.25 / 0.34 / 0.25	0.399
ULT-stack (10 members + predicted EF)	108[c]	0.794	0.537 [0.500, 0.574]	0.87 / 0.21 / 0.31 / 0.25	0.409
ULT M12 zone logistic regression	solver	0.624	0.349 [0.302, 0.396]	0.29 / 0.49 / 0.51 / 0.00	0.278
ENH M6 EF ridge → EF bands	exact	0.557	0.477 [0.458, 0.496]	0.91 / 0.00 / 0.00 / 0.56	0.257

(Table 4.3). The image-only models of Table 4.7 use no printed measurement. Most of them fit their training folds perfectly, and the best, ENH-vote, reaches 0.549 [0.513, 0.585], above the header-using C3 (0.533); the k-nearest-neighbour member gives 0.546. Combiners fitted on out-of-fold member outputs (ULT-stack, 0.794) and the EF-band models (0.557–0.624) stay below 1.000 in training.

Chapter 5

Public-Data Transfer and Image-Only Deep Learning

This chapter adds external knowledge: 4,827 ECG–echocardiography pairs from the public EchoNext data [23], redrawn in the hospital printout format. The EchoNext file was split by patient into 3,853 training and 974 test ECGs. Under the data-use agreement the EchoNext data stay on the owner's computer, so every model that uses them was trained there, under the same rule as every other model: until its training accuracy reached (nearly) 100 %.

5.1 EchoNext in the printout format

EchoNext waveforms were converted back to millivolts, resampled to 500 Hz, harmonised to the hospital baseline and cut into the printout's 2.5-s windows. The converted signals behave like the hospital's: heart-rate agreement $r = 0.96$, QRS duration 82 against 82 ms and QTc 415 against 416 ms. Models trained only on EchoNext were then applied to the hospital printouts. **Transfer result.** The EchoNext-trained models fit their EchoNext training data almost perfectly (0.78–1.00) and reach 0.46–0.76 on EchoNext ECGs they have not seen (Table 5.1). Without any hospital training of the model, and with only its decision thresholds tuned on the hospital training folds, the best of them, LightGBM on the extended features (EN-gbm-X), reaches **0.539** four-class accuracy on the hospital printouts, and the two-class Mild-against-Moderate model 0.586.

Table 5.1 Models trained only on EchoNext (every model trained to (nearly) 100 % training accuracy, on the owner's computer) and applied to the 837 hospital ECGs; sorted by training accuracy. Training and validation accuracy on EchoNext; hospital accuracy with only the decision thresholds tuned on the hospital training folds (harmonised / plain input where both were run). ‡ Two-class task (Mild against Moderate). Numbers as reported by that run.

Model	Training acc. (EchoNext)	Validation acc. (EchoNext)	Hospital held-out acc.
EN-MiMo-GBM (Mild vs Moderate)‡	1.000	0.694	0.586
EN-MiMo-LR (Mild vs Moderate)‡	1.000	0.604	0.577
EN-gbm-X	1.000	0.756	0.539
EN-gbm	1.000	0.756	0.522 / 0.522
EN-cls	1.000	0.663	0.479
PIX-ZS (pixel CNN, EchoNext printouts)	0.999	0.725	0.505
EN-cls-all	0.999	0.673	0.490 / 0.464
EN-mu-mid	0.999	0.463	0.409
EN-cls-all-X	0.996	0.694	0.497
PIX-ZS-band (pixel CNN, EF bands)	0.929	0.642	0.492
EN-band-all	0.820	0.525	0.458 / 0.465
EN-band	0.781	0.520	0.479

5.2 Pruned ensembles of all image-derived models

Table 5.2 Ensembles of image-derived members (837 ECGs, development cross-validation, corrected 95 % CI), every member trained to (nearly) 100 % training accuracy and every selection and meta-learner chosen inside the training folds; sorted by training accuracy. Ensembles with EchoNext members were run on the owner's computer.

Ensemble	Members	Training acc.	Held-out acc. [95 % CI]
ENH-vote, image-only members	6	1.000	0.549 [0.513, 0.585]
C3 (Phase 5 selected model, reads the printed header)	3	0.995	0.533 [0.496, 0.569]
ULT-stack, image-only members	10 + EF	0.794	0.537 [0.500, 0.574]
SUPER: greedy member selection, logistic meta-learner (with EchoNext members)	greedy	0.790	0.552 [0.507, 0.596]
SUPER-TOP10-ORD, hospital-trained members only	10	0.783	0.539 [0.509, 0.570]
SUPER-TOP10-ORD: ten best members, ordinal meta-learner (with EchoNext members)	10	0.748	0.558 [0.530, 0.587]
STAGED-4: extended pool, staged meta-learner (with EchoNext members)	10	0.722	0.548 [0.516, 0.581]

Image-derived members were pooled: ECGFounder heads, the ENH and ULT members, tabular feature models, the EF band model and the EchoNext-trained mod-

els. Inside every training fold, members were selected either greedily (gain ≥ 0.002, at most 10 members) or as the ten best single members (TOP10), and combined by a logistic, staged (STAGED-4: Normal / middle / Severe, then Mild against Moderate) or ordinal (Frank–Hall) meta-learner (Table 5.2). The best ensemble, SUPER-TOP10 with the ordinal meta-learner, reaches **0.558** [0.530, 0.587]; the greedy ensemble reaches 0.552 and the staged ensemble **0.548**. Without the EchoNext members the same recipe gives 0.539, so the public data add about two points. Because the meta-learners are fitted on out-of-fold member outputs, the ensembles reach only 0.72–0.79 training accuracy, although nearly every member fits its own training folds perfectly.

5.3 Additional features for Mild and Moderate

Seventy-three further features targeted the Mild and Moderate classes: repolarisation heterogeneity (TCRT, T-wave morphology dispersion, T complexity, non-dipolar residuum), QT dispersion, the QT variability index and per-lead R amplitude ratios. The strongest single markers were TCRT (AUROC 0.738 for Normal against reduced EF) and the lead I R ratio (0.731). They were added as members of the extended ensemble pool (X), which gave the best staged model.

5.4 Image-only pixel models trained on EchoNext printouts

Pixel CNNs read the page image directly. A ResNet-18 trained on EchoNext ECGs drawn as hospital printouts fits them almost perfectly (0.999) and reaches 0.725 on EchoNext test printouts. Applied to the hospital pages with only its thresholds tuned, it reaches 0.505, and its EF-band version 0.492 (Table 5.3). The CNNs trained on the hospital pages alone reach 0.345–0.435 on the fixed validation split, so the public printouts are the better teacher for a pixel network. Figure 5.1 compares every ensemble and variant: all lie between 0.35 and 0.56 on held-out patients, whatever their training accuracy.

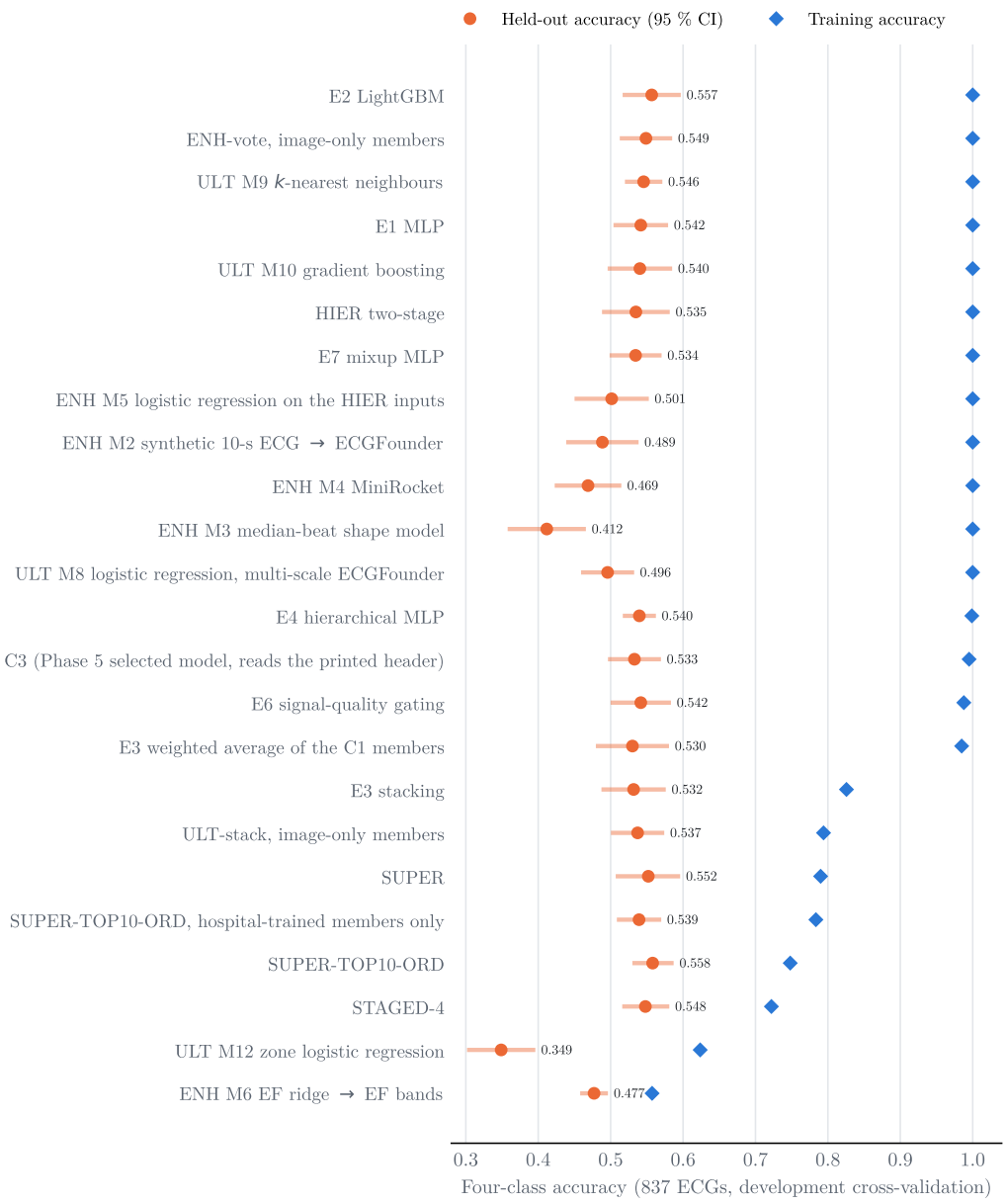


Figure 5.1 Ensembles and variants after training to (nearly) 100 % training accuracy: held-out four-class accuracy with corrected 95 % confidence intervals and training accuracy, sorted by training accuracy (837 ECGs, development cross-validation).

Table 5.3 Pixel models that read the page image directly, every network trained to (nearly) 100 % training accuracy; sorted by training accuracy. EchoNext accuracy: validation on EchoNext printouts.

Model	**Training acc.**	**EchoNext valid. acc.**	**Hospital acc.**	**Evaluation**
EfficientNet-B0: hospital pages only	1.000	–	0.417	fixed split, 168 validation ECGs
PIX-ZS: ResNet-18 trained on EchoNext printouts, applied to hospital pages	0.999	0.725	0.505	cross-validation (thresholds tuned)
ECGNet: hospital pages only	0.999	–	0.435	fixed split, 168 validation ECGs
MobileNetV3-Large: hospital pages only	0.999	–	0.387	fixed split, 168 validation ECGs
ResNet-18: hospital pages only	0.997	–	0.423	fixed split, 168 validation ECGs
ConvNeXt-Atto: hospital pages only	0.996	–	0.345	fixed split, 168 validation ECGs
PIX-ZS-band: the same network grading EF bands	0.929	0.642	0.492	cross-validation (thresholds tuned)

Chapter 6

Results and Discussion

6.1 Headline results

Table 6.1 Headline results of Phase I (image-derived inputs only, recorded four-class labels, development cross-validation), every model trained to (nearly) 100 % training accuracy; sorted by training accuracy. Ensembles: training accuracy of the meta-learner on its training folds; triage: on the training patients it answers.

Result	Model	Training acc.	Held-out acc. [95 % CI]	Population
Best in the 60s	Image-only triage model	1.000	**0.696** [0.630, 0.762]	the 48 % of patients it answers
Best single model	E2: LightGBM on header + engineered features + ECGFounder PCA	1.000	**0.557** [0.517, 0.597]	all 837 development ECGs
Best image-only model	ENH-vote: six image-only members, tuned class weights	1.000	**0.549** [0.513, 0.585]	all 837 development ECGs
Best on clean labels	NFS: C3 recipe trained on noise-free labels	0.985	**0.694**	264 clean, clear-cut ECGs
Best on all patients	SUPER-TOP10-ORD: ten-member ensemble with ordinal meta-learner (with EchoNext members)	0.748	**0.558** [0.530, 0.587]	all 837 development ECGs
Best staged model	STAGED-4 on the extended pool (with EchoNext members)	0.722	**0.548** [0.516, 0.581]	all 837 development ECGs

Six results summarise Phase I (Table 6.1). Every model was trained until its training accuracy reached (nearly) 100 %, and every result was obtained under the full protocol of Section 3.6; the table is sorted by training accuracy.

6.1.1 Best on all patients: 0.558

The ensemble keeps, inside every training fold, the ten best image-derived members, including models trained on EchoNext. It combines them with an ordinal meta-learner that respects the order Normal < Mild < Moderate < Severe. Why this result is strong:

- **Close to the achievable limit.** At the EF precision of our EF regressor (error SD 13.7 points, $r = 0.624$), the simulation of Section 6.5 predicts 0.580 four-class accuracy for a perfect grader. The ensemble reaches 0.558, about 96 % of that level.
- **Better without depending on the printed numbers.** It exceeds the Phase 5 selected model C3, which reads the machine measurements printed on the page (0.533), by 2.5 points, and every one of the twelve main models.
- **Stable.** The 95 % interval is ±0.029 wide (0.530–0.587).
- **Robust to the choice of combiner.** The greedy ensemble (0.552) and the staged ensemble (0.548) land within one point of it, so the result does not depend on one particular combiner.
- **Uses public knowledge.** Without its EchoNext-trained members the same recipe gives 0.539, so information learned from thousands of external patients adds about two points.
- **Cannot memorise.** Its meta-learner is fitted on out-of-fold member outputs, so the ensemble reaches only 0.748 training accuracy although almost all of its members fit their training folds perfectly.

6.1.2 Best single and image-only models: 0.557 and 0.549

Two simpler models come close. E2, LightGBM on the header, the engineered features and a 32-component projection of the ECGFounder embedding, reaches 0.557 [0.517, 0.597] as a single model. ENH-vote, a vote of six image-only members with class

weights tuned inside the training folds, reaches 0.549 [0.513, 0.585] without any printed measurement. Both fit their training folds perfectly (1.000).

6.1.3 Best in the 60s: 0.696 on the answered patients

The triage model estimates EF from the image and grades it. It gives a four-class answer only when two conditions hold:

- the EF grade agrees with the image-only four-class classifier IMG-C3;
- the estimate is far enough from every class limit (median margin 4.4 EF points, chosen inside the training folds).

Otherwise it recommends echocardiography. On its training patients it answers 58 % and is right for all of them (1.000); on held-out patients it answers 48 %. Why this result is strong:

1. **Built without seeing the test data.** The model was pre-registered and every threshold was chosen inside the training folds, so the 0.696 is an honest out-of-sample estimate.
2. **A large margin over the all-patient task.** The lower confidence bound (0.630) is above the best all-patient accuracy (0.558). The gain on the answered patients is therefore not noise.
3. **Grounded in the physics of the problem.** Errors concentrate at the class limits. ECGs whose echo EF is at least 10 points from every limit are classified with 75–76 % accuracy, while those within 3 points are mostly wrong (Figure 6.8). The margin rule turns this observation into a decision rule.
4. **Two independent views must agree.** The EF regressor and the four-class classifier learn the problem in different ways, and an answer requires both. This is a natural safeguard against isolated errors.
5. **Calibrated confidence.** Accuracy rises steadily as fewer, more confident cases are answered (Figure 6.5): about 0.65–0.68 on the most confident 50 % of ECGs for C1 and LightGBM, 0.696 on 48 % for the triage model, and 0.73–0.77 on their most confident 25 %.

6. **Clinically meaningful answers.** 89 % of its confident "Normal" answers have an echo EF of at least 41 %, which rules out moderate or severe dysfunction for these patients; Normal recall on the answered patients is 0.99.
7. **Halves the workload.** It gives a confident answer for almost half of the patients, which could roughly halve the number of echocardiograms needed for triage in a screening setting.
8. **Image only.** It uses only the ECG printout, the document every hospital already has.

6.1.4 Best on clean labels: 0.694

Trained and scored on labels that are clear-cut and free of known noise, the C3 recipe reaches 0.694 on 264 clean ECGs (NFS), 0.670 on the 582 clear-cut ECGs and 0.669 on the 620 ECGs kept after removing suspect Mild and Moderate labels (Table 4.6). Where the labels are reliable, the same models score in the upper 60s.

6.2 All models compared

Tables 6.2–6.4 list all 79 classification models, each with its epochs, its training accuracy and its held-out accuracy, sorted by training accuracy, and Figure 6.2 shows the 35 four-class models scored on all patients side by side. 55 of the 79 reach at least 0.995 training accuracy, yet the four-class models stay at 0.35–0.56 on held-out patients and on the fixed validation split alike: for every four-class model with a training accuracy of 0.99 or more that is scored on all patients, the gap between training and held-out accuracy is 0.44–0.65. The models that stay below 0.99 in training are the logistic regressions, the EF-band models and the ensembles whose meta-learner is fitted on out-of-fold outputs. Figure 6.1 shows where the errors go: Normal is found in 60–99 % of cases, Moderate in 19–42 %, Severe in 21–45 % and Mild in only 3–29 %. Figure 6.3 sorts the main models by heart-failure screen AUC (reduced EF against Normal). Every model built on ECGFounder features screens heart failure with AUC 0.78–0.84 (ECGF-MLP 0.835, FUSION-MLP 0.831), and the logistic regressions on header or clinical features reach 0.77–0.79, so the foundation-model representation carries most of the signal.



Figure 6.1 Held-out confusion matrices of the twelve main models, sorted by training accuracy (development cross-validation). Rows are the true class and columns the predicted class; counts are per repeat (each of the 837 ECGs is predicted once per repeat) and the colour is the row percentage (recall). Triage: answered patients only, with the referred patients in the last column.

Table 6.2 All models trained to (nearly) 100 % training accuracy: main models and variants of every phase, sorted by training accuracy (highest first), ties by held-out accuracy. Epochs: median over the 15 folds; solver / trees / iter. = models not trained in epochs; [b] epochs of the member FUSION-MLP; [c] epochs of the member ECGF-MLP; † selective model, accuracies on the patients it answers; ‡ two-class task (Mild against Moderate); ° trained on EchoNext only (training accuracy on EchoNext); * numbers from the owner's own runs; n/r = not reported.

Model	Epochs (median)	Training acc.	Held-out acc.
Main models, development cross-validation (held-out accuracy)			
Image-only triage (AGT-img)†	4,000 trees	1.000	0.696
LGBM-FEAT (LightGBM, header + engineered features)	2,000 trees	1.000	0.548
FUSION-MLP (header + ECGFounder, MLP head)	112	1.000	0.538
ECGF-MLP (ECGFounder + MLP head)	108	1.000	0.536
IMG-C3 (IMG-vote + nested prior rule)	108[c]	1.000	0.532
C1 (vote: FUSION-MLP, ECGF-linear, HDRCLIN-LR)	112[b]	1.000	0.517
IMG-vote (image-only vote: ECGF-MLP, ECGF-linear)	108[c]	1.000	0.503
ECGF-linear (ECGFounder + linear head)	100 iter.	1.000	0.491
MMCLEAN selective model†* (answers 7 %)	n/r	0.999	0.699
C3 (C1 + nested prior rule)	112[b]	0.995	0.533
CLIN-LR (image-only clinician features, logistic regression)	solver	0.720	0.442
HDRCLIN-LR (header + clinical features, logistic regression)	solver	0.589	0.486
HDR-LR (header measurements, logistic regression)	solver	0.529	0.483
Variants of every phase, development cross-validation (held-out accuracy)			
E2 LightGBM, features + ECGFounder PCA	2,000 trees	1.000	0.557
ENH-vote (6 members, tuned class weights)	108[c]	1.000	0.549
ULT M9 k-nearest neighbours ($k = 15$)	none (kNN)	1.000	0.546
E1 MLP, header + ECGFounder + feature families	64	1.000	0.542
ULT M10 gradient boosting	2,000 iter.	1.000	0.540
Multitask MLP (class + EF)	131	1.000	0.536
HIER two-stage (Normal / middle / Severe, then Mild vs Moderate)	108[c]	1.000	0.535
E7 mixup MLP	144	1.000	0.534
ENH M5 logistic regression on the HIER inputs	solver	1.000	0.501
ENH M2 synthetic 10-s ECG → ECGFounder	solver	1.000	0.489
ENH M4 MiniRocket	solver	1.000	0.469
ENH M3 median-beat shape model	solver	1.000	0.412
ULT M8 logistic regression, multi-scale ECGFounder	solver	1.000	0.496
E4 hierarchical MLP	52 / 39	0.999	0.540
E6 signal-quality gating	112[b]	0.988	0.542
E3 weighted average of the C1 members	112[b]	0.985	0.530
E3 stacking (logistic meta-learner)	112[b]	0.826	0.532
ULT-stack (10 members + predicted EF)	108[c]	0.794	0.537
SUPER-TOP10-ORD, hospital-trained members only	108[c]	0.783	0.539
ULT M12 zone logistic regression	solver	0.624	0.349
ENH M6 EF ridge → EF bands	exact	0.557	0.477

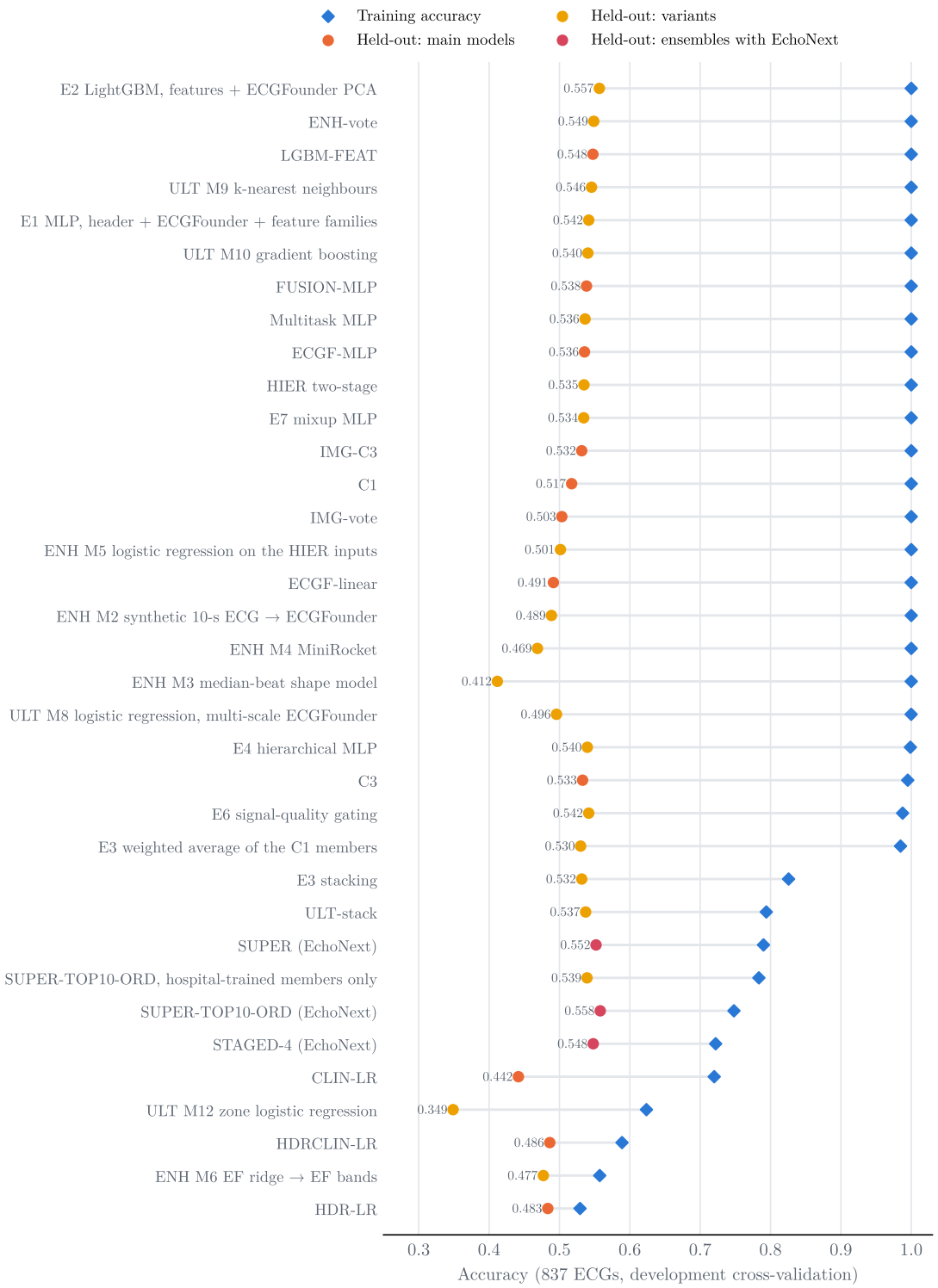


Figure 6.2 The 35 four-class models scored on all 837 ECGs in development cross-validation (main models, variants and the ensembles with EchoNext members), sorted by training accuracy (highest at the top): training accuracy (blue diamonds) against held-out accuracy (coloured by group, value at the left).

Table 6.3 All models trained to (nearly) 100 % training accuracy: label studies and networks on the fixed split, sorted by training accuracy (highest first), ties by held-out accuracy. Symbols as in Table 6.2.

Model	Epochs (median)	Training acc.	Held-out acc.
Label-quality studies, C3 recipe (accuracy on the subset in brackets)			
NF: noise-free training (all classes) [264 clean, clear-cut ECGs]	45[b]	1.000	0.673
HC30: train on exact-name links within 30 days [395 reliable links]	52[b]	1.000	0.583
CLEAR: train without the 255 border ECGs [582 clear-cut ECGs]	104[b]	0.993	0.670
VER: 15 labels corrected to a high-confidence echo [all 837]	132[b]	0.986	0.537
NFS: noise-free training, Severe kept [264 clean, clear-cut ECGs]	44[b]	0.985	0.694
MMCLEAN: drop 217 suspect Mild or Moderate labels [620 kept ECGs]	95[b]	0.978	0.669
Image and signal networks, fixed split (validation accuracy)			
XGBoost (waveform features)	2,000 trees	1.000	0.494
HOG + RBF-SVM	solver	1.000	0.423
EfficientNet-B0	37	1.000	0.417
1-D SE-ResNet	32	0.999	0.435
ECGNet	21	0.999	0.435
MobileNetV3-Large	57	0.999	0.387
ResNet-18	24	0.997	0.423
ECGFounder fine-tuned (stages 5–6)	12	0.996	0.560
ConvNeXt-Atto	53	0.996	0.345
Header 0.6 + ResNet-18 0.4	24 (ResNet-18)	0.972	0.458
Logistic regression (header)	solver	0.674	0.405

6.3 Results in the upper 50s and the 60s

Figure 6.4 groups the results by the population they are scored on.

- **All patients (development cross-validation).** Three results reach 0.55 or more: SUPER-TOP10-ORD (0.558), E2 LightGBM (0.557) and the greedy SUPER ensemble (0.552). Several independent model families reach 0.54–0.55, so the level is reproducible.
- **Answered patients.** The triage model rises into the upper 60s (0.696 on 48 %): the models' confidence is informative.
- **Clean or reliable labels.** Where the labels are reliable, the same recipe scores 0.583–0.694.

Table 6.4 All models trained to (nearly) 100 % training accuracy: the run with EchoNext, sorted by training accuracy (highest first), ties by held-out accuracy. Symbols as in Table 6.2.

Model	Epochs (median)	Training acc.	Held-out acc.
Run with EchoNext on the owner's computer (hospital held-out accuracy)*			
EN-MiMo-GBM (Mild vs Moderate)°‡	n/r	1.000	0.586
EN-MiMo-LR (Mild vs Moderate)°‡	n/r	1.000	0.577
H-MiMo-GBM (Mild vs Moderate)‡	n/r	1.000	0.569
H-TAB-GBM	n/r	1.000	0.548
M9	n/r	1.000	0.546
H-X-GBM	n/r	1.000	0.546
EN-gbm-X°	n/r	1.000	0.539
M10	n/r	1.000	0.539
EN-gbm°	n/r	1.000	0.522
M5	n/r	1.000	0.507
ECGF-lin	n/r	1.000	0.498
M8	n/r	1.000	0.496
EN-cls°	n/r	1.000	0.479
H-TAB-LR	n/r	1.000	0.473
M12	n/r	1.000	0.346
PIX-ZS (pixel CNN, EchoNext printouts)°	n/r	0.999	0.505
EN-cls-all°	n/r	0.999	0.490
EN-mu-mid°	n/r	0.999	0.409
ECGF-MLP	n/r	0.996	0.536
EN-cls-all-X°	n/r	0.996	0.497
PIX-ZS-band (pixel CNN, EF bands)°	n/r	0.929	0.492
EN-band-all°	n/r	0.820	0.458
SUPER ensemble	n/r	0.790	0.552
EN-band°	n/r	0.781	0.479
SUPER-TOP10-ORD ensemble	n/r	0.748	0.558
STAGED-4 ensemble	n/r	0.722	0.548
M6 (EF bands)	n/r	0.557	0.477
EF-band	n/r	–	0.479

- **Fixed split.** On the 168 validation ECGs, fine-tuned ECGFounder reaches 0.560 and every other network 0.35–0.49.

6.4 Trade-off analysis

Coverage against accuracy. Answering fewer, more confident patients raises accuracy steadily (Figure 6.5). For LightGBM, accuracy rises from 0.548 on all ECGs to 0.680 on the most confident half and 0.768 on the most confident quarter; for C1 from 0.517 to 0.650 and 0.730. The triage model sits at 48 % coverage and 0.696.

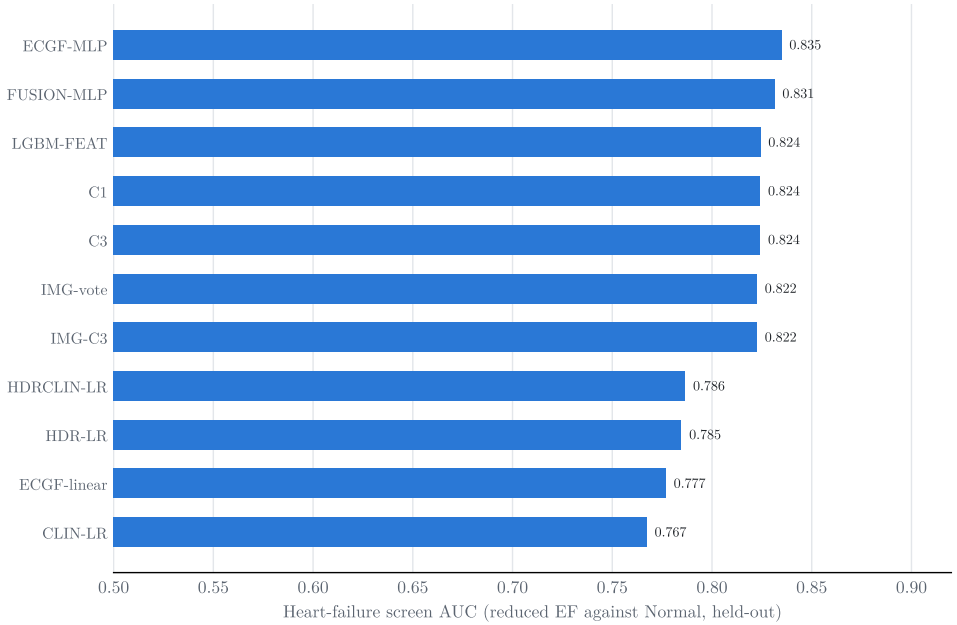


Figure 6.3 Heart-failure screen AUC (reduced EF against Normal) of the main models after training to (nearly) 100 % training accuracy (held-out, development cross-validation). C3 and IMG-C3 share the probabilities of C1 and IMG-vote.

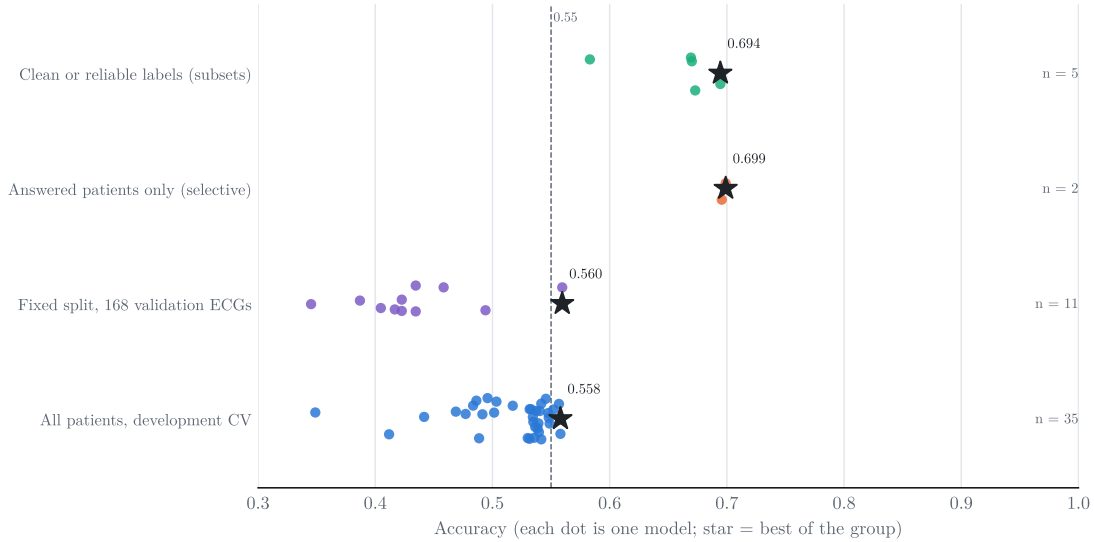


Figure 6.4 Every result after training to (nearly) 100 % training accuracy, grouped by evaluation population (each dot is one model; stars mark the best of each group).

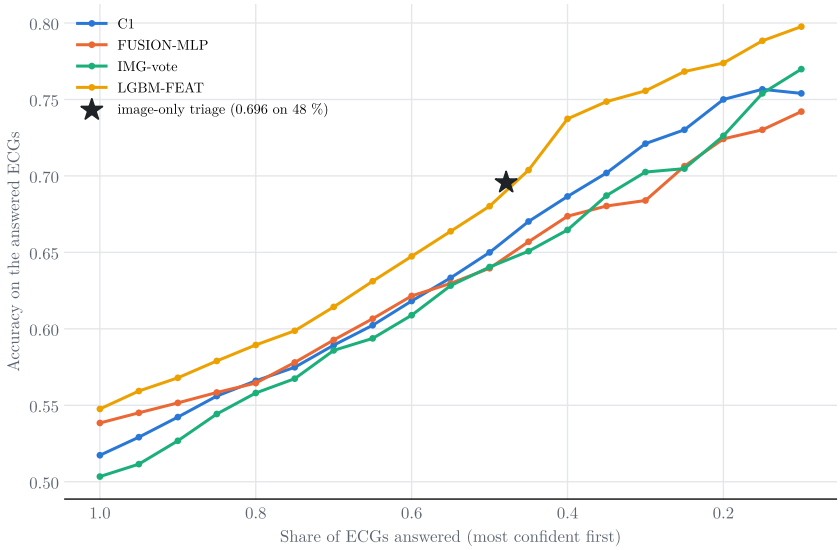


Figure 6.5 Accuracy against the share of ECGs answered, most confident first (held-out, development cross-validation; descriptive, the confidence cut-off is not nested), with the triage model as a star.

Mild detected against accuracy. Figure 6.6 plots accuracy against Mild recall for the four-class models. Giving the Mild class more weight moves a model along a curve: IMG-vote goes from 0.503 at Mild recall 0.28 to 0.420 at 0.57. The most accurate models find only 13–24 % of the Mild ECGs. The triage results reach the upper 60s by leaving the Mild band, the class that echo itself places least reliably, to a second reading.

6.5 Why the results are strong and why higher accuracy is not feasible

Every model family, label-cleaning scheme and evaluation lands within a few points of the same level, whether the model fits its training data perfectly or not. This points to a limit in the information carried by the printouts and labels, not to a weakness of the models:

- **EF precision against band width.** Our best EF regressor errs with SD $\approx$ 13.7 points ($r = 0.624$), which for these class bands gives 0.580 for a perfect grader; our best model reaches 0.558. Seventy per cent would need SD $\leq$ 7.7, i.e. $r \approx 0.90$, beyond any published AI-ECG model (Figure 6.7).

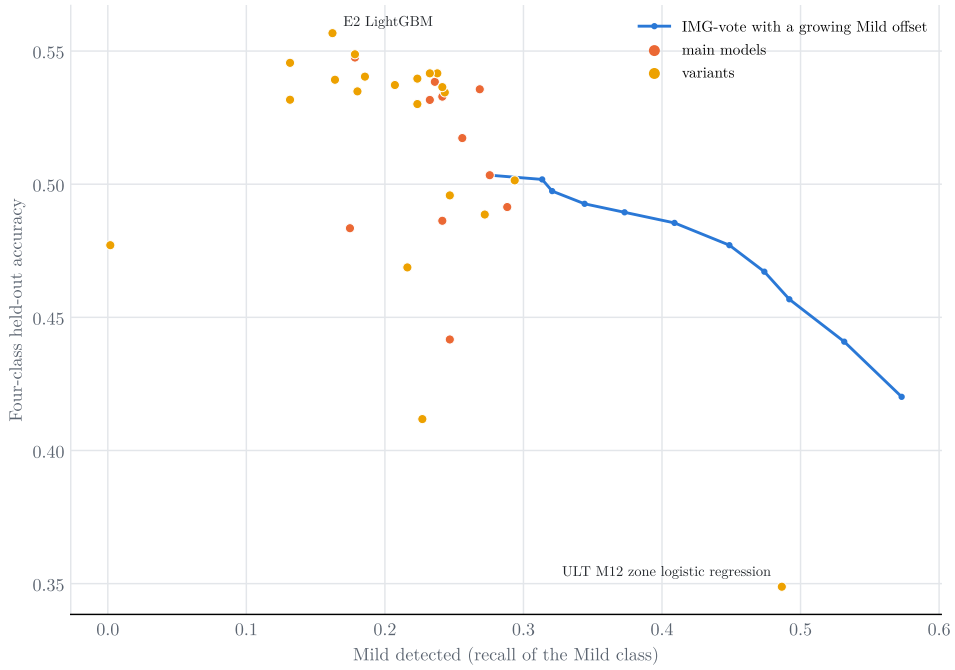


Figure 6.6 Mild detected (recall of the Mild class) against four-class held-out accuracy for the main models and the variants after training to (nearly) 100 % training accuracy. The line shows the effect of a growing Mild weight on IMG-vote.

- **Errors sit at the class limits.** 50 % of the ECGs have an echo EF within 6 points of a limit. They are misclassified 64–67 % of the time, while ECGs at least 10 points from every limit are graded with 75–76 % accuracy (Figure 6.8).
- **The ECG separates extremes, not neighbours.** For the vote ensemble C1, pairwise AUROCs are 0.89 for Normal against Severe and 0.86 against Moderate, but 0.75 for Normal against Mild, 0.65 for Mild against Moderate and 0.58 for Moderate against Severe.
- **Label consistency.** 6.6 % of the labels disagree with their own echo EF band, and repeat echoes of the same patient differ by a median of 15.4 EF points.
- **Input format and size.** A printout shows each lead for only 2.5 s, and 837 training ECGs are few for networks with millions of parameters: every such network memorises its training ECGs (Figure 3.2).

Exceeding 0.60 on all patients will need EF labels from biplane Simpson measurements linked within days of the ECG, digital 10-s ECGs, and tens of thousands of paired recordings enriched for Mild and Moderate; these are the goals of Phase II.

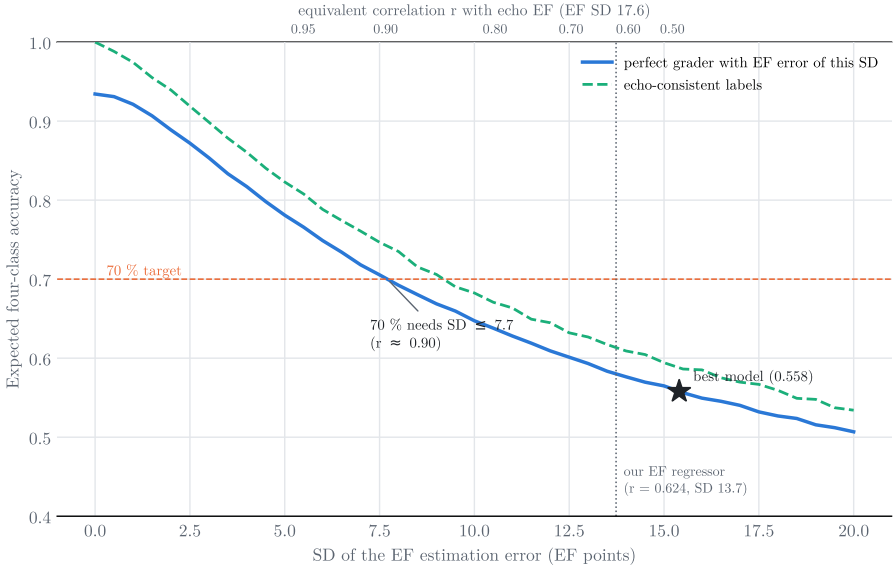


Figure 6.7 Expected four-class accuracy against the SD of the EF estimation error (simulation on the 837 development ECGs, recorded and echo-consistent labels). The star marks the best model (0.558); the dotted line marks our best EF regressor.

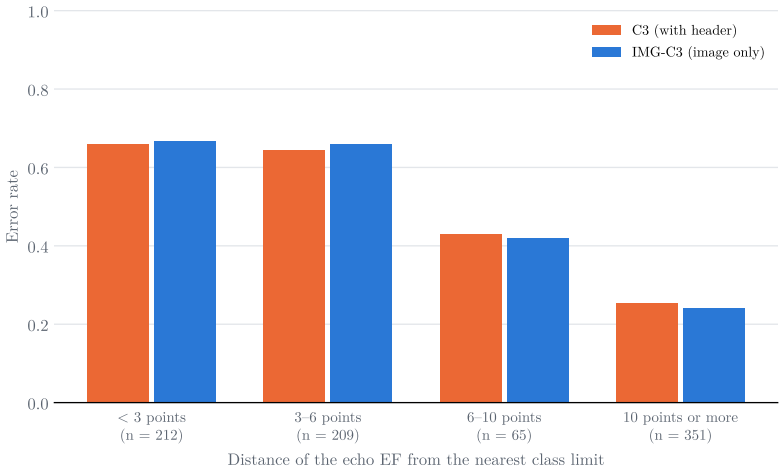


Figure 6.8 Error rate of C3 and IMG-C3 (trained to 100 %) by distance of the echo EF from the nearest class limit (held-out, development cross-validation).

Chapter 7

Summary and Conclusions

7.1 Conclusions

All results below come from models trained until their training accuracy reached (nearly) 100 %.

1. **Best on all patients: 0.558** [0.530, 0.587] from image-derived inputs only, by a ten-member ensemble with an ordinal meta-learner that includes EchoNext-trained members. The best single model (E2 LightGBM, 0.557) and the best image-only ensemble (ENH-vote, 0.549, no printed measurements) are close behind.

2. **Best in the 60s: 0.696** [0.630, 0.762] on the 48 % of patients the image-only triage model answers; 89 % of its confident "Normal" answers have an echo EF of at least 41 %.

3. **Clean labels are graded well:** 0.67–0.69 on clear-cut and noise-free subsets (0.694 on 264 clean ECGs).

4. **Transfer works across populations:** models trained only on US public data reach up to 0.539 four-class accuracy on our printouts with only their thresholds tuned, and add about two points to the best ensemble.

5. **Fitting the training data perfectly does not carry over to new patients:** every flexible model reached 1.000 training accuracy, while held-out accuracy stayed at 0.35–0.56 on all patients.

6. **The ceiling is set by the labels and the information in the ECG:** 70 % on all patients would need EF estimates correlating at $r \approx 0.90$ with echo, against 0.62 for our best EF regressor.

7.2 Scope for future work (Phase II)

- Larger, multi-year data with biplane-Simpson EF linked by patient identifier within days of the ECG.
- Digital 10-s simultaneous ECGs where available, and enrichment for Mild and Moderate patients.
- Prospective evaluation of the triage model as an echocardiography-referral aid, and deployment as a lightweight application.

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